Phuc Nguyen Tran Thien

Nguyễn Trần Thiên Phúc · AI researcher · M.Sc. student in Computer Science

Ho Chi Minh City, Vietnam · phucntt.258i@ou.edu.vn

Portfolio / Google Scholar / ORCID / GitHub / LinkedIn

Research interests: visual speech recognition, sign language recognition, multimodal learning, reproducible model evaluation, and AI for scientific sensing. Seeking international research assistantships, research collaborations, and MSc/PhD opportunities.

EDUCATION

Ho Chi Minh City Open University Aug 2025–Present

M.Sc. in Computer Science (Artificial Intelligence), in progress; expected 2027.

FPT University, Ho Chi Minh City 2021–2025

Bachelor's degree in Artificial Intelligence.

RESEARCH EXPERIENCE

Research Assistant · Ho Chi Minh City Open University Aug 2025–Present

• Develop computer vision and multimodal research across Vietnamese sign language, visual speech recognition, and smartphone chemical sensing.

• Design experiments, audit dataset partitions, train models, analyze errors, and create reproducible evaluation and inference workflows.

• Co-author six publications, including two first-author IEEE JCSSE papers and an article in *Analytica Chimica Acta*.

SELECTED RESEARCH

Visual speech recognition and temporal scope · 2026

• Study the relationship between visual context, text supervision, and output scope; contribute to matched-frame continuation experiments and interval/complete-transcript evaluation.

• Release research code and numerical replay artifacts for *Decoupling Visual Context and Text Scope in Visual Speech Recognition* (research manuscript). Code and manuscript

Vietnamese sign language and sign–lip fusion · 2025–2026

• Develop isolated-sign recognition and multimodal fusion pipelines, with pose/skeleton and lip features, model comparison, and exported-prediction audits.

• Build streaming recognition and assistive-interface prototypes; document camera and recording-group limitations when interpreting multimodal results. Research archive

Smartphone AI for chemical sensing · 2025–2026

• Investigate calibration and normalization for ammonium/nitrite colorimetric strips; develop ROI extraction, multi-task classification/regression, and a FastAPI inference service. Code

INDUSTRY EXPERIENCE

AI Engineer Intern · FPT Software Sep–Dec 2024

Built OCR and semantic-search components using EasyOCR, Sentence Transformers, Elasticsearch and MongoDB; integrated hybrid BM25/dense retrieval, metadata filters and REST APIs with application teams.

PUBLICATIONS

Six published papers · two first-author papers. Author names follow publication records.

1. **Phuc Nguyen Tran Thien**, Trung Nguyen Quoc, Thai-Binh Tran, Vinh Truong Hoang, Kiet Tran Trung (2026). **Deep Smartphone-Based Classification of NH4+ and NH2- Colorimetric Strips: Revisiting the Role of Explicit Green-Border Calibration**.
IEEE JCSSE 2026 · First author. doi:10.1109/jcsse68839.2026.11597064

2. **Phuc Nguyen Tran Thien**, Trung Nguyen Quoc, Thanh-Danh Nguyen, Vinh Truong Hoang, Kiet Tran-Trung (2026). **Deep NH4+ Regression for Smartphone Colorimetric Test Strips: On the Need for Explicit Green-Border Normalization**.
IEEE JCSSE 2026 · First author. doi:10.1109/jcsse68839.2026.11597039

3. Trung Nguyen Quoc, **Phuc Nguyen Tran Thien**, Vinh Truong Hoang, Kiet Tran-Trung (2026). **Adapting Hierarchical Feature Aggregation and Spatio-Temporal Graph Neural Networks for Vietnamese Visual Speech Recognition on a Small Phrase-Level Dataset**.
IEEE CICN 2026. doi:10.1109/cicn70047.2026.11594397

4. Trung Nguyen Quoc, **Phuc Nguyen Tran Thien**, Vinh Truong Hoang, Kiet Tran-Trung (2026). **Toward Edge-Deployable Vietnamese Lip Reading: A Compact Lipnet-Style Pytorch Recognizer with a Jetson-Oriented Inference Pipeline**.
IEEE CICN 2026. doi:10.1109/cicn70047.2026.11594266

5. Trung Nguyen Quoc, Minh-Vuong Phan, Khoi Pham Dang, Huy Ngo Huu Gia, **Phuc Nguyen Tran Thien**, et al. (2026). **Smartphone-based colorimetric sensing with reference calibration and ensemble machine learning for enhanced detection of nitrite and ammonium ions**.
Analytica Chimica Acta, 1388, 345097. doi:10.1016/j.aca.2026.345097

6. Trương Duy Việt, Ngô Hữu Gia Huy, Phạm Đăng Khôi, **Nguyễn Trần Thiên Phúc** (2025). **MoViNet-A2 cho bài toán nhận diện ký hiệu Tiếng Việt**.
HCMCOU Journal of Science - Engineering and Technology, 20(2), 3–14. doi:10.46223/hcmcoujs.tech.vi.20.2.4201.2025

SELECTED RESEARCH SOFTWARE

LipTrace — Dataset audits, bound CTC model bundles, PyTorch/ONNX trace comparison and corpus replay.

VSL Streaming Toolkit — Python library, CLI and WebSocket service for ordered keypoints and isolated-sign segment events.

Pose Reference Toolkit — Reference banks, temporal pose alignment and inspectable geometric feedback.

METHODS AND TECHNICAL SKILLS

Research: experiment design; dataset and split audits; ablation studies; error analysis; CTC decoding; cross-dataset evaluation; reproducible numerical analysis.
Tools: Python, PyTorch, OpenCV, MediaPipe, ONNX Runtime, scikit-learn, FastAPI, Docker, Git, Linux, Flutter.

SELECTED AWARDS

1st Place, Cardano SEA Hackathon (AI Track).
Third Prize, Food Innovation and Development 2025, Food Security & Social Impact category (student team lead).
Top 10, Southeast Asia Sparki Startup Competition; Top 5, Youpreneur Launchpad 2024; Top 10, I-Startup 2024.